

PE03: Programming Exercise

The concept of Structured Query Language (SQL) was developed to communicate with relational databases. It is the American National Standards Institute (ANSI) standard for viewing, adding, updating and deleting data in a relational database management system (DBMS). Some examples of types of DBMS systems are: Oracle SQL, Microsoft SQL Server, IBM UDB, Sybase, MySQL and Informix. Research SQL concepts at the below link and keep them in mind for this exercise.

<https://dev.mysql.com/doc/refman/8.0/en/select.html>

Your task for Module 03 is to:

1. Use ANSI SQL to extract and view data from a relational database table using aggregate functions.
2. Review the following contents of table SCUBA_DIVER (Table 1.0).
3. Write a SQL statement to do the following:
 - a. Select data from the SCUBA_DIVER table to display all SCUBA diver information by last name, first name in descending order. Use the Alias Names for each column name, "SCUBA ID", "Diver Last Name", "Diver First Name", "Certification No.", "Certification Date", "Certification Code", "Instructor No.", "Certification Level" group by Certification Level.
 - b. Select data from the SCUBA_DIVER table to display all SCUBA diver information and group by Instructor Id. Use the Alias Names stated above.
 - c. Select data from the SCUBA_DIVER table to count the number of Scuba Divers. Use an Alias Name as stated above to describe the resulting column.
 - d. Select data from the SCUBA_DIVER table to display distinct values of each Instructor Id.
 - e. Select data from the SCUBA_DIVER table to display distinct values of each Certification Level.
 - f. Select data from the SCUBA_DIVER table to display all SCUBA diver information for the SCUBA diver(s) with certifications less than 6 months old group by Instructor Id.
 - g. Select data from the SCUBA_DIVER table to display all SCUBA diver information for the SCUBA diver(s) with the oldest certification date group by Instructor Id.
 - h. Select data from the SCUBA_DIVER table to display all SCUBA diver information for the SCUBA diver(s) with the most recent certification date group by Instructor Id.
 - i. Select data from the SCUBA_DIVER table and display the unique types of SCUBA diver certification levels.
 - j. Select data from the SCUBA_DIVER table and display the lowest SCUBA Id in the table.
 - k. Select data from the SCUBA_DIVER table and display the highest SCUBA Id in the table.

4. What is a constraint? What about this feature improves the efficiency of data stored in a database? Justify your answer and results for input that will be used in the future PE(s).

SCUBA_DIVER_ID	LAST_NAME	FIRST_NAME	CERTIFICATION_NUMBER	CERTIFICATION_DATE	CERTIFICATION_CODE	INSTRUCTOR_ID	CERTIFICATION_LEVEL
1	HOFFMAN	DONNA	9905492628	1999-05-05	P	1876	OPEN WATER
2	SMITH	HARRY	8774533967	1990-04-20	P	3340	OPEN WATER
3	TAYLOR	HENRY	7776000011	2001-01-20	P	7511	ADVANCED OPEN WATER
4	LEWIS	VALERIE	5509087633	2004-08-15	P	6992	SCUBA
5	HARRISON	GARY	3299940002	2004-08-20	H	6992	HAZWOPER/HAZMAT
6	FOSTER	TERRY	3887000555	2006-06-22	H	7511	HAZWOPER/HAZMAT
7	HAWTHORNE	MARY	8776000999	2007-04-20	ADCI	7511	ADCI COMMERCIAL
8	JENNINGS	BEATRICE	8342223668	2005-01-01	P	4110	RESCUE
9	FORD	WILLIAM	8674199800	2005-03-17	DAI	4110	DIVER ACADEMY INTERNATIONAL
10	NELSON	ISAAC	6540077717	2021-01-15	P	9870	MASTER
11	WILLIAMS	CINDY	6740009981	2021-11-27	P	4110	WRECK
12	WILLIAMS	EDWARD	9340227596	2014-07-13	P	4110	ADVANCED OPEN WATER
13	RICHARDS	SCOTT	3887000555	2016-11-08	P	3340	ENRICHED AIR
14	ADAMS	PHILLIP	5663210905	2010-02-05	DAI	6992	DIVER ACADEMY INTERNATIONAL
15	MUNSON	REX	4779805522	2019-08-23	ADCI	9870	ADCI COMMERCIAL
16	JONES	JEFF	6441229005	2019-09-17	DAI	6992	DIVER ACADEMY INTERNATIONAL
17	HOLLY	EVAN	9800600437	2017-02-28	ADCI	3031	ADCI COMMERCIAL
18	WILCOX	AMY	8099499512	2021-05-12	H	3031	HAZWOPER/HAZMAT
19	LEE	BETTY	9902111144	2011-10-08	ACDE	9870	ACDE COMMERCIAL
20	HOWARD	VICTORIA	9444432876	2012-12-12	ACDE	1876	ACDE COMMERCIAL
21	IVY	MAPLE	7777443299	2021-06-01	P	3031	RESCUE

Table 1 (SCUBA_DIVER)